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Module : **Computer System and Networks 8x8Arena**

1. Project Description

8x8Arena is crowd-controlled mini-game platform inspired by the classic "Twitch Plays" experiment. A small game like (a Snake-style grid game) runs on the LED matrix of a Raspberry Pi SenseHat , while remote players send a command for movement from their phones or browser over the network. All inputs are published to a central message broker, processed in real time on the Pi, and the resulting game state is reflected both in the physical LED and on a live web dashboard. The goal is to demonstrate a full IoT pipeline – sensor/input layer, processing, networking and application – within a playful, observable context.

2. Architecture

phone and browser clients send commands to an MQTT broker (Mosquitto running on the Raspberry Pi), which forwards them to a Python game loop. The game loop processes the inputs and outputs the result to both the SenseHAT 8×8 LED matrix and a live web dashboard.

3. How will Works

Input layer - Can be player remotely using a simple web page on their phones and tap directional buttons. The local SenseHat joystick also acts as a fallback player.

Networking layer – Each input is published as a JSON message e.g. :
(`{"player": "p1", "cmd": "up"}`) to an MQTT topic running on a Mosquitto broker hosted on the Raspberry Pi.

Processing layer – A Python game-loop on the Pi subscribes to the command topic, applies for the rules: movement, collisions, scoring and updates the game state at a fixed tick rate.
Application layer – The current state is rendered on the SenseHAT 8X8 LED matrix in real time, and also pushed to a web dashboard that show live game state, player activity and basic stats.

4. Tools, Technologies & Equipment

Hardware

Raspberry Pi (with SenseHAT — 8×8 LED matrix and joystick used)
Raspberry Pi Camera Module 3 (live stream of the LED matrix in the dashboard) Players' personal devices (phone/laptop browser) as remote controllers

Software & Languages

Python 3 — game logic and MQTT client (`paho-mqtt`, `sense-hat` libraries)
HTML / CSS / JavaScript — web dashboard and remote controller page (using MQTT.js over WebSockets)

Protocols & Services

MQTT (Mosquitto) for publish/subscribe messaging
HTTP for serving the static web dashboard

Tools

VS Code — development
Git & GitHub — version control and submission
Mosquitto — MQTT broker

5. Project Repository

https://github.com/nrossidasilva1/8x8Arena